SC - GitHub Copilot 2

Duration: 10 minutes.

* Indicates required question

1. Email *

2. Full name *

3. Student ID *

4. In the context of applying acquired prompt crafting skills in practical settings, which of the following actions demonstrates effective application and evaluation? (Select all that apply.) 1 point

Check all that apply.

- [ ] Analyzing Copilot's responses to refine the specificity and clarity of subsequent prompts.
- [ ] Applying few-shot learning by including multiple related examples within the prompt to guide Copilot's output.
- [ ] Critiquing prompts based on their alignment with the 4S criteria and revising them for improved effectiveness.
- [ ] Relying solely on zero-shot techniques for all types of prompts due to their simplicit

5. How does GitHub Copilot handle ambiguous or incomplete prompts when generating code suggestions? (Select all that apply.) 1 point

Check all that apply.

- [ ] It always asks for clarification.
- [ ] It makes the best guess based on available data.
- [ ] It generates multiple options for consideration.
- [ ] It does not generate any code.

6. In the context of GitHub Copilot, what is the significance of "few-shot learning", and how does it differ from "zero-shot" and "one-shot" learning in practice? 1 point

Mark only one oval.

- ( ) Few-shot learning uses several examples, unlike zero-shot, which uses none and one-shot, which uses one example.
- ( ) Few-shot learning is more accurate than both zero-shot and one-shot learning.
- ( ) Few-shot learning, zero-shot, and one-shot learning are all similar in practice.
- ( ) Few-shot learning is only effective for complex prompts, unlike zero-shot and one-shot.

7. In which scenarios might GitHub Copilot's suggestions be suboptimal or incorrect? (Select all that apply.) 1 point

Check all that apply.

- [ ] When the suggestion does not follow best coding practices.
- [ ] If the suggestion is based on outdated libraries.
- [ ] When the suggestion is not aligned with project-specific guidelines.
- [ ] When the suggestion overlooks the latest industry standards.

8. When integrating GitHub Copilot into diverse coding scenarios, which of the following is a true statement about its utility? 1 point

Mark only one oval.

- GitHub Copilot can only generate code snippets for web development projects.
- GitHub Copilot assists by suggesting code snippets across various programming languages and frameworks, adapting to different coding scenarios.
- GitHub Copilot requires manual input of the entire code before it can suggest improvements.

9. In the context of applying GitHub Copilot in a practical project, how can it streamline real-world coding tasks? 1 point

Mark only one oval.

- By automatically writing the entire project code without any developer input.
- By exclusively focusing on generating documentation for the code instead of the code itself.
- By providing code suggestions and optimizations that reduce the time spent on routine coding tasks.

10. While working on a web application, you ask GitHub Copilot to generate code for user login functionality. Copilot suggests code that introduces a security vulnerability (SQL injection). How would you refine your prompt to ensure Copilot generates a more secure and contextually appropriate code suggestion? 1 point

Check all that apply.

- [ ] Trust the initial code suggestion provided by Copilot, assuming it meets security standards.
- [ ] Adjust your prompt to emphasize the importance of secure coding practices, such as implementing parameterized queries for database interactions.
- [ ] Use a broad prompt like "Generate a login function" without including specific security requirements.
- [ ] Enhance the prompt by including details about sanitizing user input and avoiding vulnerabilities like SQL injection.

11. What is a key best practice for using GitHub Copilot to ensure efficiency and maintain code quality? 1 point

Mark only one oval.

- Always accept the first code suggestion provided by GitHub Copilot.
- Review and understand the code suggested by Copilot before integrating it.
- Use GitHub Copilot only for writing boilerplate code.
- Rely solely on GitHub Copilot for all coding tasks.

12. Which of the following is NOT an editable setting in GitHub Copilot? 1 point

Mark only one oval.

- Keyboard shortcuts configuration.
- Language preferences for code suggestions.
- Use a broad prompt like "Generate a login function" without including specific security requirements.
- Automatically include security measures like input sanitization and protection against vulnerabilities such as SQL injection in code suggestions.

13. You are using GitHub Copilot to help develop a new feature for your project, which requires integrating a third-party library that you are not familiar with. How should you use Copilot to efficiently learn how to implement this library while ensuring the generated code aligns with your project's standards? 1 point

Mark only one oval.

- Generate code snippets with Copilot and incorporate them into your project without additional checks.
- Use Copilot to create example code by explaining what you need from the library, then carefully review the output to understand how the library functions.
- Let Copilot manage the code generation for the library, bypassing manual testing since the AI should produce correct code.
- Turn off Copilot when working with new libraries to avoid potential errors in your codebase.